WEEK 4 · DAY 1 · INSTRUCTOR ONLY

Quiz 2 Answer Key

Week 3 · shuffled version

Instructor only. Built from the original `Time-series/MCQ/week-04-mcq.csv` (that file is unchanged), with 3 questions edited: see *Changes from the original* below. Options were reshuffled with a fixed seed (4104) so the correct letters are spread out: multiple-choice answers sit at A ×7, B ×7, C ×6, D ×6 (in the original, most answers were option B). The 4 True/False questions keep their original order.

#	Answer	Correct option	What it tests	Level
1	**A**	True	A threshold is a model	beginner
2	**D**	Every alarm is technically correct but operationally useless	Seasonal false alarms	beginner
3	**A**	Zero	Silent degradation	beginner
4	**D**	Spike, level shift and drift	Three anomaly shapes	beginner
5	**D**	It is a response to rain, so it leaks the answer	Leaking the answer	advanced
6	**A**	Where did that number come from?	Where thresholds come from	intermediate
7	**C**	Tampering	Common versus special cause	intermediate
8	**C**	Three standard deviations either side of the reference-period mean	Control chart parts	beginner
9	**B**	About 8	Three sigma as a budget	intermediate
10	**B**	They are a different operating state and would widen the limits	Reference period	intermediate
11	**D**	Level shifts	Run rule 2	intermediate
12	**B**	Fit a separate set of limits for each operating state	One chart per state	beginner
13	**A**	Observed value minus expected value	Predict, subtract, chart	beginner
14	**C**	A few extreme points cannot move them	Median and MAD	intermediate
15	**C**	To make it comparable to a standard deviation	The 1.4826 factor	advanced
16	**A**	True	The anomaly hides in its baseline	intermediate

#	Answer	Correct option	What it tests	Level
17	**B**	Slow drift	What residuals miss	intermediate
18	**A**	Values all above zero	Is the residual clean	beginner
19	**A**	A level shift	Shift versus drift	beginner
20	**B**	Small deviations from target, so a sustained change becomes a clear slope	CUSUM idea	beginner
21	**C**	It is slack, so ordinary noise does not accumulate	The slack k	intermediate
22	**B**	False	Alarm date versus onset	intermediate
23	**B**	The rolling slope of capacity	Battery fade knee	intermediate
24	**C**	No single reading looks extreme against the whole series	TWTA level shift	advanced
25	**A**	About 99.87%	Why not accuracy	beginner
26	**A**	Of the real events, how many we caught	Recall	beginner
27	**D**	Precision	Precision	beginner
28	**D**	As one catch	Event-level scoring	intermediate
29	**B**	It turns a statistical choice into a decision an operations manager can make	Pricing false alarms	intermediate
30	**B**	False	Unusual is not wrong	advanced

Changes from the original

The same fixes are in the weekly question banks (`week-N/MCQ/`).

- **Q2** (corrected): It said the floor fires on "all 46 eclipse hours". On the data it fires 18 times, all during eclipses.
- **Q4** (reworded): Week 4 of the original course is Week 3 here.
- **Q23** (corrected): It said the fade rate "roughly doubled". The data shows -2.24 to -3.68 Ah a year: 1.6x faster.

Marking it together (10 minutes)

1. Read out the letters; everyone marks their own paper.
2. Ask for hands up on each question. Any question that more than a third of the room got wrong, explain it again with the matching day's slide.
3. Map each topic back to its day: what your alarm software already does → W3D1 · statistical process control → W3D2 · residual-based detection → W3D2 · change points and drift → W3D3 · evaluating a detector → W4D1.

Quick topic tally

Topic	Questions
what your alarm software already does	1, 2, 3, 4, 5, 6
statistical process control	7, 8, 9, 10, 11, 12
residual-based detection	13, 14, 15, 16, 17, 18
change points and drift	19, 20, 21, 22, 23, 24
evaluating a detector	25, 26, 27, 28, 29, 30